

Environment-Limited Synchronization: Distributed Arrays That Phase-Lock on Ordinary Traffic

Chandler Stevenson

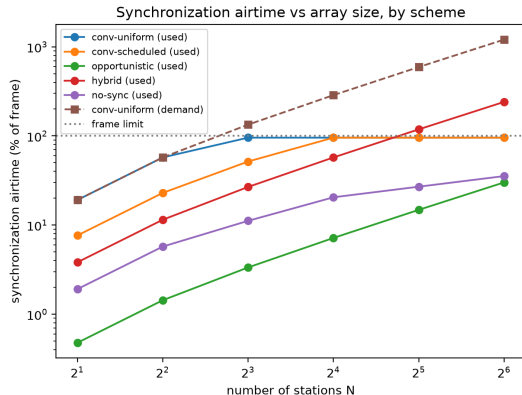
Princeton University

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Waveform-level simulation study — full oscillator and RF impairments; no hardware yet.

The problem: synchronization eats the channel

- A distributed transmit array only achieves coherent beamforming gain while every station's carrier phase is locked to the others.
- Conventional over-the-air locking uses *dedicated two-way exchanges*: stations take turns sending known signals; comparing the two directions cancels the propagation path and measures the clock offset alone.
- That cost is per-station, per-interval. As the array grows, synchronization traffic saturates the frame: measured collapse at $N \approx 8$ (naive schedule) and incoherence by $N=64$ even for the strongest conventional baseline.



Synchronization airtime vs. array size, per scheme.

The conventional design rule — and the claim of this work

The field's rule: synchronization rate is priced by *oscillator quality*.

- Update-interval analyses derive re-synchronization cadence from oscillator and PLL phase noise (Mghabghab & Nanzer, *IEEE Access* 2021).
- BeamSync, AirSync, and the TDD-integrated calibration line all transmit *synchronization-specific* signals — references, preambles, or beamformed sync waveforms — at a cadence set by clock drift.

This work inverts the rule:

When ordinary traffic carries the tracking, the required rate of dedicated two-way exchanges is set by the *environment's* coherence time — not by the clocks.

The idea: ordinary transmissions as free phase observations

- The array radiates data and sensing bursts constantly. Every time station j overhears station i 's transmission, it observes a phase — at *zero marginal airtime*.
- Closest ancestor: Quitin *et al.* (2013) extract frequency/phase from the modulated waveform of *explicitly generated feedback packets*, referenced to a designated receiver's clock.
- Delta here: **no synchronization-specific transmission exists at all**. Arbitrary payload-bearing bursts between array elements are the observations, and the transmitting stations' own oscillators are steered in closed loop.

measured: phase extracted from unmodified random-data OFDM bursts is *as good or better* than a dedicated preamble (4.78 vs. 8.75 mrad per observation — slide 12 explains why).

The catch: an impossibility result

A one-way observation mixes two things that must not be confused:

$$y_{ij} = \underbrace{\theta_i - \theta_j}_{\text{clock offset}} + \underbrace{\psi_{ij}}_{\text{propagation phase}} + n.$$

Track the per-link state $x = [\theta, \omega, \psi]$ (clock phase, clock frequency, path phase). One-way observations have measurement row $H_{\text{free}} = [1, 0, 1]$: they see only the *sum*.

Non-identifiability (proved)

The observability Gramian of (F, H_{free}) has rank 2. Its null direction is exactly $(1, 0, -1)/\sqrt{2}$: **no amount of one-way data — at any signal-to-noise ratio, over any duration — can separate clock from path.**

Ablation confirms the stakes: a 2-state filter that ignores the path state steers the transmitter by the propagation phase — 1327 ± 100 mrad, beam quality collapses to 62.6%.

The fix: sparse dedicated two-way exchanges

- A dedicated two-way exchange measures the clock offset *alone*: the half-difference of the two directions cancels the (reciprocal) path, $\frac{1}{2} \text{wrap}(\varphi_{ij} - \varphi_{ji}) \rightarrow \theta_i - \theta_j$. Its measurement row $[1, 0, 0]$ restores the Gramian to full rank.
- One such exchange every K update intervals re-pins the clock/path decomposition; free observations track the clocks in between.

The residual π ambiguity, resolved with one bit

The half-difference observes $\text{wrap}(2\theta)$ — a 2-to-1 map, so exactly two candidates $\{\hat{\theta}, \hat{\theta} + \pi\}$ remain, and the wrong one is an attractor (zero innovation while the transmitters cancel). A periodic one-bit test, $\text{sign}(\cos \theta_{\text{err}})$, decides a two-element set: one bit is *necessary and sufficient*. Measured under adverse acquisition (initial offset 2.2 rad): **12/12 seeds lock anti-phase without the check, 0/12 with it.**

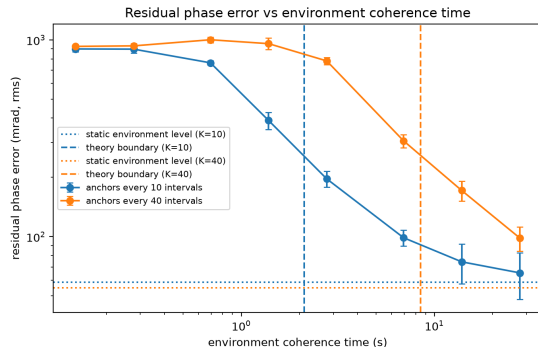
When does the decomposition survive? A zero-fit boundary

Environment motion makes the path phase drift; the filter's gains misattribute the deterministic Doppler ramp to the clocks. The locked bias after K intervals is $\approx \pi f_D T K$, giving the identifiability condition

$$\pi f_D T K < b$$

(f_D Doppler frequency, T update interval, K exchange spacing, b phase budget). **No oscillator parameter appears.**

- Predicted collapse: 0.066 m/s ($K=10$), 0.016 m/s ($K=40$) at $b = 0.314$ rad.
- Measured: within one grid step of both, no fitted constants; three regimes (free / biased / broken) with the predicted signatures.



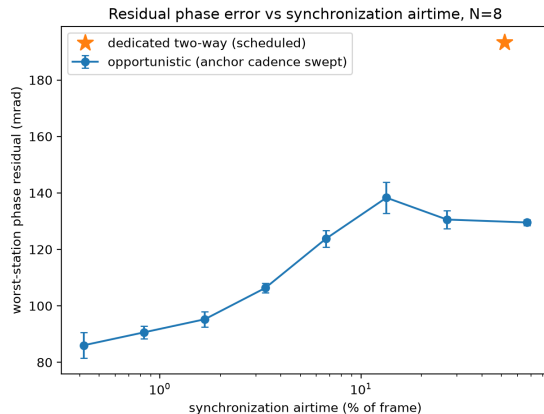
Residual vs. environment coherence time, two exchange cadences; dashed verticals = predicted boundaries.

Result 1: rarer dedicated exchanges *improve* accuracy

Sweep the exchange spacing $K = 2 \rightarrow 320$ at $N = 8$, static environment:

- 129.5 mrad at 66.9% synchronization airtime $\rightarrow$ **86.0 ± 4.5 mrad at 0.42%.**
- Airtime cut 160 $\times$; the residual does not degrade — it improves, because each two-way capture itself injects multipath measurement noise.
- Every point on the frontier beats the dedicated baseline (193.3 mrad at 51.6%) on *both* axes.

Static-environment best case; sparsity is bounded by the slide-7 condition, not by clock drift.



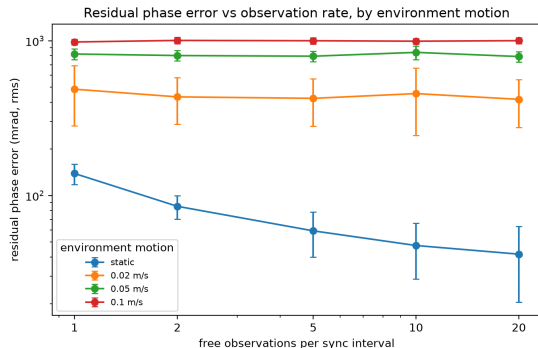
Residual vs. dedicated-exchange airtime.

Result 2: free observations never hurt (hypothesis test)

An expert reviewer predicted a U-shaped error curve in a slowly varying environment: too many free observations should let channel motion masquerade as clock drift. Our theory predicted no U.

- 5 seeds, 4 environment speeds, observation rate $1 \rightarrow 20$ per interval, post-convergence statistics.
- Static: monotone improvement, $138.7 \rightarrow 41.7$ mrad.
- Moving: *flat* — the locked bias is independent of observation rate (290–327 mrad over a $20\times$ rate range at 0.02 m/s).

Verdict: no U-curve at any speed — the real tradeoff lives on the exchange-spacing axis (slide 7).

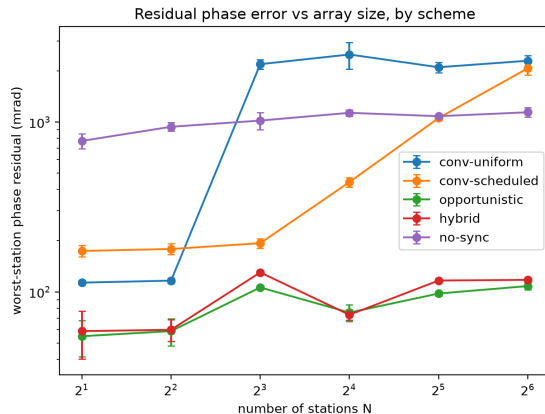


Residual vs. free-observation rate, per environment speed.

Result 3: flat accuracy from 2 to 64 stations

N	opportunistic	conv. (scheduled)	conv. (uniform)
2	55 mrad / 0.5%	174 mrad / 7.6%	113 mrad / 19.1%
8	106 mrad / 3.3%	193 mrad / 51.6%	2190 mrad / 95.6%
16	75 mrad / 7.2%	443 mrad / 95.6%	2498 mrad / 95.6%
64	108 mrad / 30.1%	2072 mrad / 95.6%	2294 mrad / 95.6%

- Opportunistic: 55 $\rightarrow$ 108 mrad over a $32\times$ size change, beam quality $\geq 99.7\%$ throughout.
- A dense-exchange control ($K=5$) matches the accuracy but its exchange demand alone exceeds the frame at $N \geq 32$: the saving comes from *replacing* dedicated exchanges with free observations, not from the estimator.

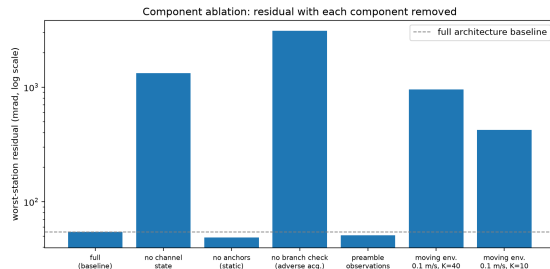


Residual vs. array size, per scheme (log scale).

Ablations: which components are load-bearing

configuration	residual
full architecture	54.7 ± 13.2 mrad
no path state (2-state filter)	1327 ± 100 mrad
no branch check (adverse acq.)	3103 ± 8 mrad
preamble instead of OFDM	51.0 ± 16.7 mrad
no exchanges after acq. (static)	40–62 mrad

- Load-bearing: the clock/path decomposition and the branch check. Waveform: irrelevant at loop level.
- Static-environment surprise: realized decomposition drift runs $\sim 1000\times$ below the worst-case bound — a static channel never excites the unobservable direction. Dedicated exchanges then serve acquisition and branch resolution; under motion they are the only defense.



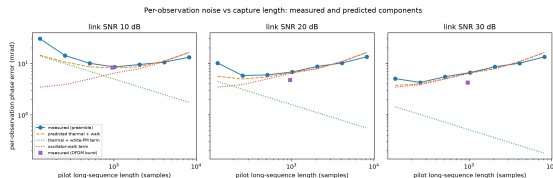
Residual per ablated configuration.

Why the data burst beats the dedicated preamble

Zero-fit model of per-observation error vs. capture length L :

$$\sigma_{\text{obs}}^2(L) \approx \underbrace{\frac{1/(2 \text{ SNR}) + \sigma_{\text{wpm}}^2}{L}}_{\text{thermal + white phase noise}} + \underbrace{\text{walk}(L)}_{\text{intra-capture oscillator random walk}}$$

- The U-shape is real at every tested SNR; predictions track measurement within 10–25% for $L \geq 511$.
- The 960-sample OFDM burst sits at the predicted optimum (4.25 predicted, 4.78 mrad measured, 20 dB); the stock preamble spans 4606 samples — the walk-dominated branch (8.75 mrad).
- A length-optimized preamble closes most of the gap (5.83 mrad): **capture geometry, not waveform magic.**



Per-observation error vs. capture length: measured points, predicted components (dashed).

Limits, and where this sits in the literature

Scope. Simulation only: 915 MHz, 1 MHz bandwidth, multipath fading with ray-traced variants, path loss, timing jitter, four-term oscillator noise anchored to real datasheets, IQ imbalance, quantization, amplifier clipping, actuation latency. The frontier numbers are the static-environment best case; the slide-7 condition is the honest boundary.

closest prior work	the gap
Quitin <i>et al.</i> 2013	sync from <i>explicitly generated</i> feedback packets, referenced to one receiver's clock; no clock/path separation problem arises
Ngo & Larsson (arXiv:2504.11411)	dedicated sync signal in engineered TDD overlap; inter-station channel <i>assumed known</i> , so the ambiguity does not exist in their model
Joint clock/channel estimation (Springer 2026)	dedicated pilots, receive-side; <i>assumes</i> the separation is identifiable, with no conditions given
Mghabghab & Nanzer 2021	re-sync interval priced by oscillator quality — the design rule this work inverts

Synchronization cadence is a property of the environment,
not of the hardware.

- ① **86 mrad** residual at **0.42%** synchronization airtime — vs. 193 mrad at 52% for dedicated two-way exchange — and rarer exchanges *improve* accuracy.
- ② Accuracy flat from **2 to 64 stations** (55–108 mrad, $\geq 99.7\%$ coherent gain) while every dedicated-signaling scheme tested is incoherent or frame-saturated at 64.
- ③ The one thing that forces a dedicated exchange is environment motion, and the boundary is closed-form: $\pi f_D TK < b$ — validated with zero fitted constants.

All numbers: `phase_sync_idea/` experiment records (seeds 0–2; hypothesis test at 5 seeds). Simulation only; hardware validation is the natural next step.